

Speed, Distance & Acceleration

Answer sheet · 27 marks total

Section A — Formula triangles

A1. Top cell = **s** (distance) Bottom-left = **v** (speed) Bottom-right = **t** (time)

[1 mark each, 3 marks total]

A2. Top cell = Δv (change in velocity) Bottom-left = **a** (acceleration) Bottom-right = **t** (time)

[1 mark each, 3 marks total]

Section B — Straight substitution

B1. $s = vt = 15 \times 8$ (1) $s = \mathbf{120\ m}$ (1)

B2. $\Delta v = at = 5 \times 3$ (1) $\Delta v = \mathbf{15\ m/s}$ (1)

Section C — Rearrange and solve

C1. $v = s / t$ (1) $v = 180 / 12$ (1) $v = \mathbf{15\ m/s}$ (1)

C2. $t = s / v$ (1) $t = 90 / 6$ (1) $t = \mathbf{15\ s}$ (1)

C3. $\Delta v = 20 - 0 = 20\ \text{m/s}$ (1) $a = \Delta v / t = 20 / 4$ (1) $a = \mathbf{5\ m/s^2}$ (1)

Section D — Scalar or vector?

(a) distance — **scalar** (1) (b) displacement — **vector** (1) (c) speed — **scalar** (1) (d) velocity — **vector** (1)

Section E — Graph interpretation

(a) First 10 s: moving at a constant (steady) speed (1). Next 5 s: stationary / at rest, no distance covered (1). Last 10 s: moving at a constant speed, greater than the first part (1).

(b) The **last part** (final 10 s) is fastest, because it has the steepest gradient on the distance–time graph (1).

Marking note: award method marks for a correctly rearranged equation and correct substitution even if the final answer contains an arithmetic error. In Section E, accept any equivalent correct description of the motion.